

salt-marsh harvest mouse 123
 Salticidae 590
Salticus 590
Saltuarius cornutus 419
 saltwater crocodile 434, 435
Salvelinus alpinus 503
 sambar 225
 San Lucas skink 423
 sand cat 202
 sand devil 482
 sand dollar 602, 604
 sand flies 568
 sand fox 170
 sand lizard 421
 grassland habitats 40
 sand monitor 430
 sand snake, cross-marked 373
 sand tiger shark 477
 sand wasp 575
 sandfish, beaked 497
 sandgrouse 23, **313**
 Burchell's 313
 crowned 313
 desert habitats 44
 Namaqua 313
 sandpipers 308
 seashore life 78
 spotted 310
 sandy seashores 76, 78
 sap-sucking insects 534
Sapajus paella 143
 saprophagous feeding 547
 sapsucker, yellow-bellied 335
Sarcophaga melanura 568
 Sarcophagidae 568
Sarcophilus harrisi 96
Sarcophyton 538
 Sarcophygyii 24, 488
 sardines 494, 495
Sardinops sagax caeruleus 495
 Sargasso Sea 491, 492
 sargassumfish 528
 satellite tracking, conservation 34
 satin bowerbird 348
 Satomi's pygmy seahorse 522
 saturniid moths 572
 Saturniidae 572
Sauromalus obesis 416
 savanna habitat 41, 42, 43
 savanna monitor 430
 Savi's shrew 163
 saw-scaled vipers 403
 sawfish,
 smalltooth 485
 wide 485
 sawflies 25, **573**
 common 575
 coniferous forest 58
 figwort 575
 sawshark, Japanese 476
 Say's harvestman 588
Sayornis phoebe 346
 scale insects 559
 scales 27
 ctenoid 513
 fish 468, 469
 placoid 473
 reptiles 372
 scallops 595
 scaly tailed flying squirrels 132
 Scandentia 22, 133
Scaphiopus couchii 451
 Scaphopoda 598
 scarab beetles 564
 sacred 564
 Scarabaeidae 564
 scarlet-chested sunbird 365
 scarlet ibis 289
 scarlet macaw 341, 338
 scavengers 15, 43
Sceloporus occidentalis 416
 scent communication 28
 grassland animals 42
 mammals 89, 90
Schistocerca gregaria 552
Schistosoma mansonii 541
 Schneider's leaf-nosed bat 159
 schooling fishes 486, 487
 herrings 494
 Schreiber's green lizard 422
 Schultze's pit viper 385
Sciaena umbra 524
 scimitar-horned oryx 237
 desert habitats 44
Scinax ruber 459
 Scincomorphoidea 24
 scissor-tailed flycatcher 346

Sciurognathi 116
 Sciuromorpha 22, 116, 118–121
 Sciurus
 carolinensis 120
 vulgaris 120
 sclerites 546
Sclerorchilus rubecula 345
 Scolecophidia 24, 407
 Scolepomorpha 24, 509
 scolex 541
Scolia procera 575
 Scoliidae 575
Scolopendra hardwickei 578
 Scolopendridae 578
 scolopendrids 578
Scomber scombrus 522
Scophthalmus maximus 519
 scops owls 323
Scopus umbretta 289
Scorpaena porcus 527
 Scorpiones 587
 scorpionflies 25, 565
 common 565
 Scorpionidae 587
 scorpionids 587
 scorpions 547, 586, **587**
 common yellow 587
 false 587
 imperial 587
 reproduction 534
 scoter, white-winged 273
Scotopelia peli 317
 screamers 270
 horned 271
 scrubbirds 348
 rufous 348
 scrubwren, white browed 349
 scutes 374
Scutisorex somereni 164
 scuttle flies 568
Scyliorhinus canicula 481
 Scyphozoa 539
 scythebills 344
Scytodes thoracica 592
 Scytodidae 592
 sea anemones 79, 537, 538
 radial symmetry 26, 533
 reproduction 30
 seashore habitat 78
 sea apple cucumber 604
 sea catfish 500
 sea cucumbers 602, **604**
 sea fans 538
 sea gooseberries 74, 544
 sea hare 596
 sea lamprey 472
 sea levels, rising 33
 sea lilies 602
 sea lions 22, 90, **182**
 California 183
 New Zealand 183
 social groups 91
 South American 182, 183
 Steller's 183
 sea mats 544
 sea mouse 542
 sea otter 193, 195
 aquatic mammals 90
 seashore habitat 79
 sea pens 538
 sea slugs 597
 sea snail, violet 74, 596
 sea snakes 77, 79, 399
 sea-spiders 25, **585**
 nymphonid 585
 yellow-kneed 585
 sea squirts 14, **605**
 colonies 535
 sea stars 602
 sea urchins 602, **604**
 feeding 535
 reproduction 534
 seashore habitat 78
 seabirds,
 oceanic 75
 seashore 76, 78
 seadragon,
 common 517
 leafy 517
 weedy 522
 seahorses 471, 513
 long-snouted 517
 Satomi's pygmy 522
 seals 21, **182**
 Antarctic fur 35, 183
 aquatic mammals 90
 Baikal 185

seals *continued*
 birth 31
 brown fur 182, 183
 classification scheme 21
 common 185
 crabeater 38, 184
 eared 182
 earless 22
 ears 89
 elephant 91, 166, 182, 185
 fur 22, 182
 gray 185
 harbor 90, 185
 harp 185
 hooded 185
 leopard 65, 184
 Mediterranean monk 184
 northern fur 183
 polar regions 64, 65, 66
 seashores 77, 79
 social groups 91
 southern elephant 185
 temperature control 89
 Weddell 66, 184
 seashore habitat 76
 seasons,
 grasslands 40, 41
 temperate forests 52, 54
 tropical forests 48, 49
 sebaceous glands 90
 Secernentea 543
 secernenteans 543
Sechellophryne gardineri 452
 secretary bird 295, 296
 seed shrimps 580
 seedsnipes 309
 segmented worms 25, **542**
 segments
 arthropod 546
 insect 548
 sei whale 249
 seladang 236
Selenochlamys ysbryda 597
Semibalanus balanoides 580
Semnopithecus entellus 149
 sengis 21, **104**
 black and rufous 104
 golden-rumped 104
 gray-faced 104
 Karoo rock 104
 rufus 104
 senses and sense organs 26, 27
 amphibians 438
 birds 265
 carnivores 166
 cartilaginous fish 473
 cats 200
 dogs 168
 fish 269, 486
 hoofed mammals 210
 invertebrates 533
 reptiles 373
 rodents 116
 whales 244
Sepia officinalis 599
 Serendib scops owl 317
Sericomyia silentis 568
Sericornis frontalis 349
 Seriema 23, 302
 seriema, red-legged 302
Serinus canaria 368
 Serpentes 24, 384–407
 serrated hinge-back tortoise 382
 serval 200, 202
 Sesiidae 572
Setifer setosus 106
Setonix brachyurus 102
Setophaga
 citrina 369
 coronata 369
 kirtlandii 369
 sexual reproduction 30
 invertebrates 534
 Seychelles frog, Gardiner's 452
 shad, American 495
 shag, imperial 290
 shallow water habitats 72
 shark sucker 517
 sharks 14, 24, 473, **474**
 basking 480
 blacktip reef 482
 blue 474, 482
 bluntnose six-gill 475
 bramble 475
 bull 481
 conservation 482
 cookie-cutter 474

sharks *continued*
 crocodile 480
 eggs 31
 evolution 16
 filter feeding 74
 frilled 475
 goblin 477
 great white 479
 hammerhead 473
 lemon 473, 482
 leopard 481
 mackerel 480
 megamouth 480
 nurse 477
 ocean habitats 72, 73
 Port Jackson 476
 sand tiger 477
 sharptooth 482
 snapper 480
 spined pygmy 475
 thresher 480
 tiger 481
 velvet belly lantern 475
 walking 476
 whale 477
 whitetip reef 482
 zebra 477
 sharp-nosed bat 158
 sharp-nosed grass frog 462
 sharp-ribbed salamander 444
 sharpbill 347
 sharptooth shark 482
 shearwater, Manx 281
 sheefish 502
 sheep 230
 Anatolian 242
 Barbary 242
 bighorn 60, 242, 243
 domestic 242
 great blue 242
 mountain habitats 63
 shelduck, common 272
 shell,
 cone 596
 elephant tusk 598
 harp 596
 helmet 596
 jingle 595
 lamp 545
 top 597
 tulip 597
 tusk 598
 shells 26
 bird eggs 267
 bivalves 594
 gastropods 594
 invertebrates 532
 mollusks 594
 reptile eggs 373
 snail 26
 tortoise/turtle 374
 shelled animals,
 in acid oceans 33
 water type 68
 shield bugs 560
 shingleback 424
Shinisaurus crocodilurus 430
 ship rat 127
 shipworms 595
 common 595
 shoebill 21, 286, 289
 shore crab, common 31, 581
 short-beaked common dolphin 258
 short-beaked echidna 92
 short-eared dog 170
 short-eared owl 320
 short-finned pilot whale 259
 short-nosed bandicoot 89
 short-nosed echidna 92
 short-tailed pygmy tyrant 342, 346
 shortfin mako 480
 shovelnose guitarfish 484
 shrew opossums 21
 shrews 22, **163**
 armored 164
 bicolored white-toothed 164
 Etruscan 163
 Eurasian water 163
 forest musk 164
 giant Mexican 163
 hero 164
 life span 30
 northern short-tailed 163
 piebald 164
 pygmy white-toothed 163
 Savi's 163
 temperate forest 55

shrews *continued*
 Tibetan water 164
 tree 22, **133**
 shrike-tit, crested 351
 shrikes 343
 bush 350
 crimson-breasted 350
 southern fiscal 354
 true 354
 shrimps 581
 brine 71, 579
 burrowing 581
 clam 579
 common 581
 cypridid mussel 580
 cypridinid 580
 desert 47
 mantis 581
 mussel 580
 opossum 584
 seed 580
 skeleton 584
 tadpole 47, 579
Sialia sialis 362
 Sialidae 561
Sialis 561
 siamang 151
 Siamese algae-eater 498
 Siberian bear 180
 Siberian crane 303
 Siberian tiger 205
 Siberian tit 59
 sibia,
 black-capped 359
 rufous 359
 Sicariidae 592
 sicklebill, white-tipped 323
 side-necked turtles 374
 side-striped jackal 171
 sidewinder 406
 sidewinding 47, 385, 402, 406
 sifaka, Verreaux's 136, 139
Siganus vulpinus 526
Sigmodon hispidus 124
 sign-bearing toadlet 452
 sika 226
 silk moths 570
 silky anteater 112
Silpha americana 564
 Silphidae 564
Silurus glanis 499
 silverfish 25, 547, **550**
 urban habitat 81
 silversides 75, 513
 silvery gibbon 152
 silvery marmoset 145
 Simien jackal 172
 Simuliidae 568
Simulium 568
 Siphonuridae 551
Siphonurus lactustris 551
 siphon 595
 Siphonaptera 25, 565
 siphonophores 537
Siphonops annulatus 447
 Sipuncula 25, 544
 siren 440
 greater 442
 northern dwarf 442
Siren lacertina 442
 Sirenia 21, 90, 107
 Siricidae 575
 Sitta
 canadensis 361
 europaea 361
 Sittellus 351
 varied 351
 Sittidae 361
 six-eyed crab spiders 592
 six-gill shark 475
 skaters, pond 559
 skates 24, 470, 473, **483**
 blue 483
 skeleton shrimps 584
 skeletons 14, 15, 26, 27
 amphibians 438
 birds 264
 bony fish 486
 fish 469
 invertebrates 532, 533
 mammal evolution 88
 reptiles 372
 snakes 384
 tortoise/turtle 374
 skimmers 308, 309
 African 312
 common 551